

Analysis group meeting

Mathieu Ouillon (Mississippi State University)

03 February, 2025



MISSISSIPPI STATE
UNIVERSITY™

Cooking for Pass0v9

Cooking for Pass0v9, started Saturday 31 January.

Name	Complete	Jobs
rgd-rqa-pass0v9_018316-18316	92.00%	25
rgd-rqa-pass0v9_CuSn-18347x301	92.92%	7525
rgd-rqa-pass0v9_ShEmRa_solid-18351x29	95.35%	667
rgd-rqa-pass0v9_ShEmRa_LD2-18305x15	94.93%	375
rgd-rqa-pass0v9_CxC-18339x125	98.14%	3125
rgd-rqa-pass0v9_LD2-18307x147	92.54%	3675

CuSn, his is still not finished
Same for other workflows

recon: 5882/6020 = 97.708%

his: 135/301 = 44.850%

ana: 547/602 = 90.864%

anamrg: 236/301 = 78.405%

anaclean: 192/301 = 63.787%

I have skimmed the pass0v8;

I need to use the jput command to save it in /cache/hallb/scratch/.

Should I delete the recon files from pass0v8 now?

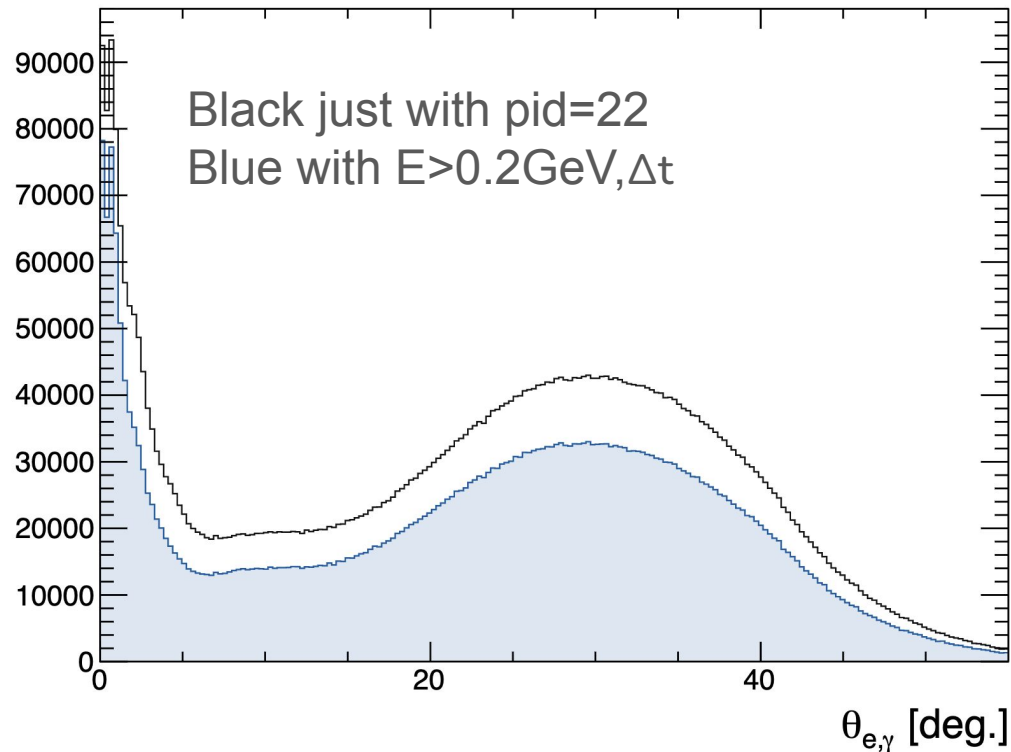
I have applied a cut on the photon to have a different sector than the electron; this cut aims to reduce the radiative photon.

A better way to reduce this is to cut on the opening angle between the electron and photon.

Data: pass0v7 LD2 data

Cuts: Electron pid=11, status<0 and
 $-11 < v_z < 0$ cm

Photons: pid =22, $E > 0.2$ GeV and $-1.67 < \Delta t < 1.55$ ns

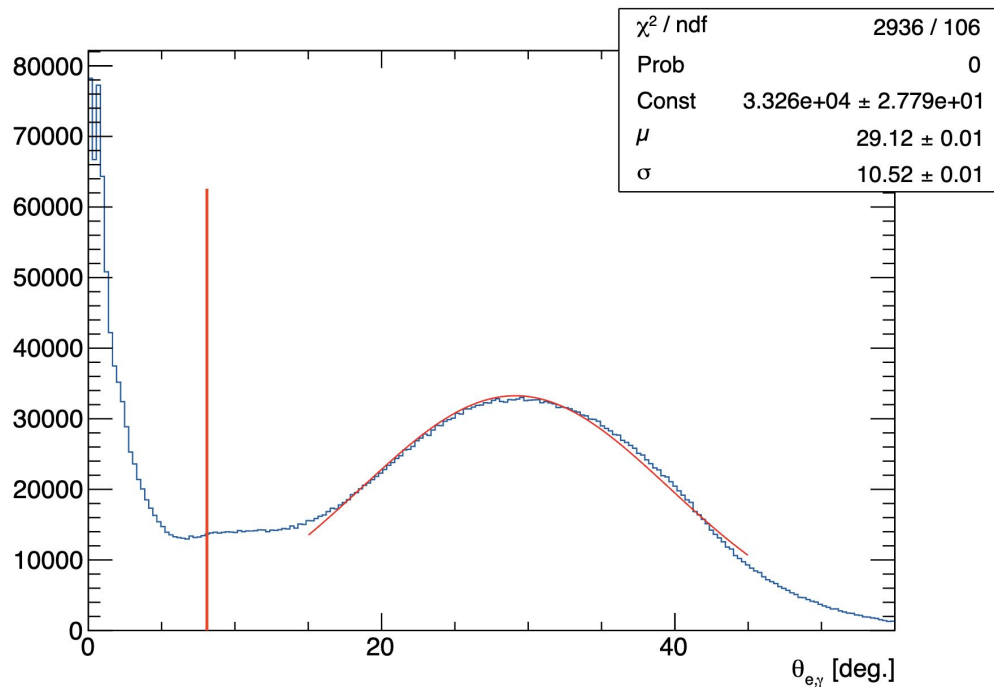


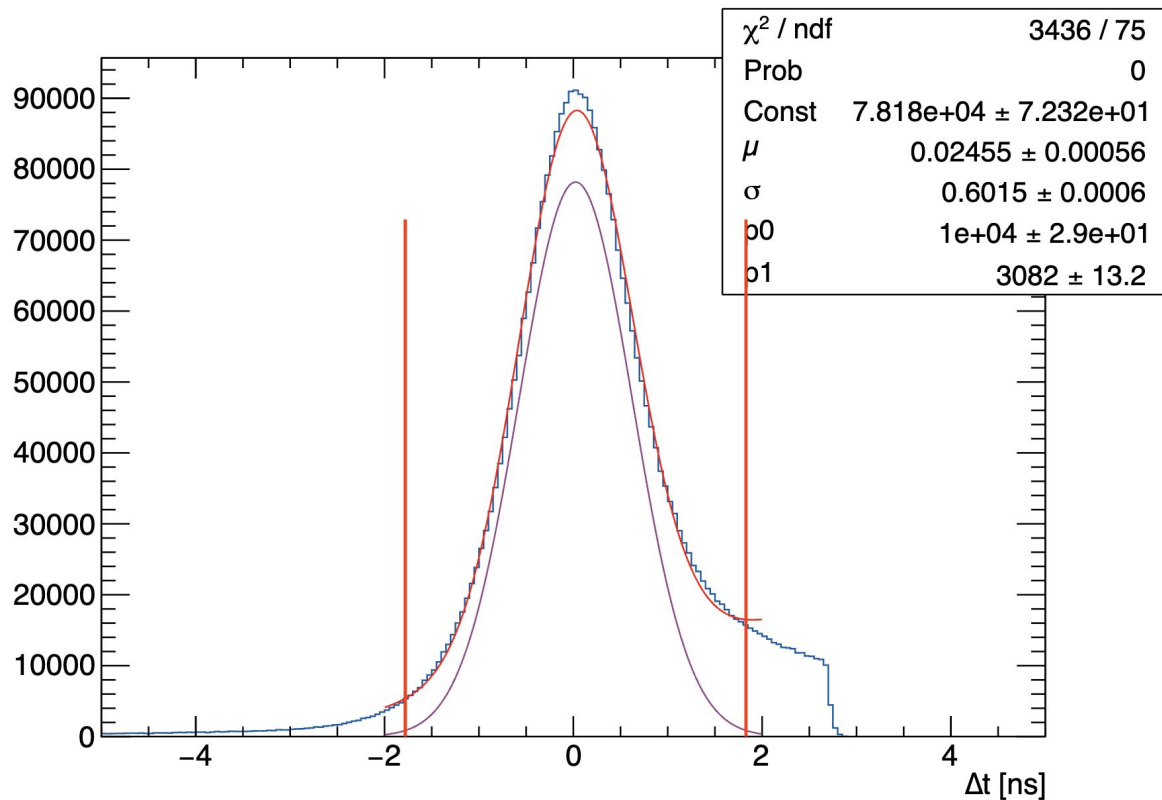
Data: pass0v7 LD2 data

Cuts: Electron pid=11, status<0 and
 $-11 < v_z < 0$ cm

Photons: pid =22, $E > 0.2$ GeV and $-1.67 < \Delta t < 1.55$ ns

Use a 2σ cut: $29.12 - 2 * 10.52$ deg





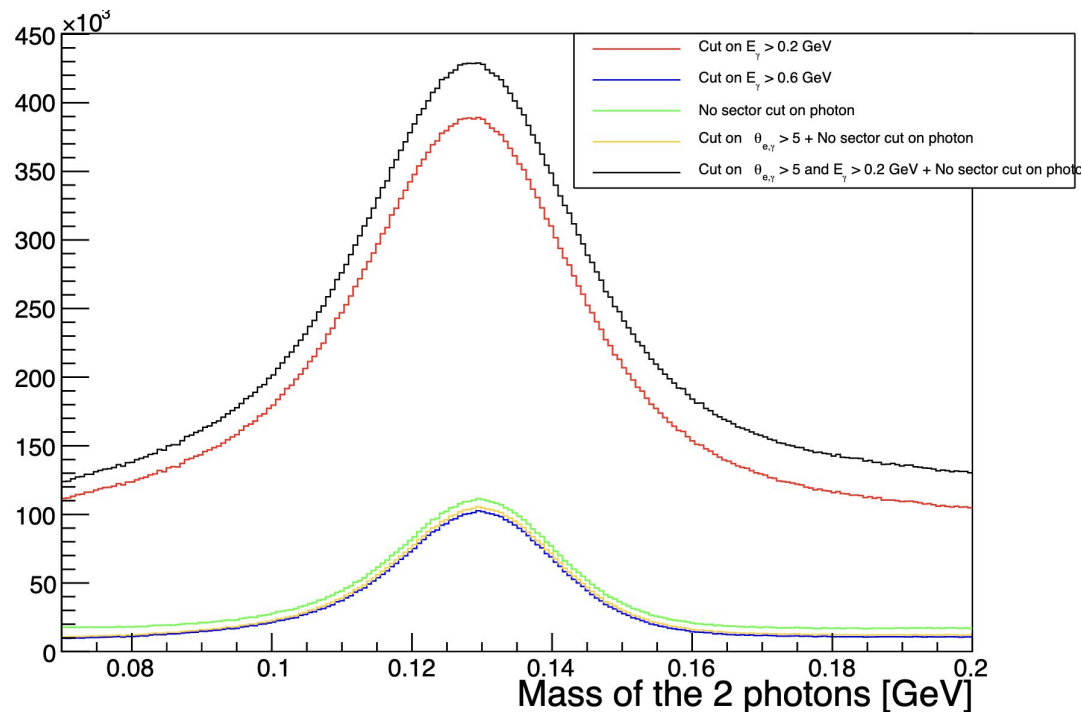
Data: pass0v7 LD2 data

Cuts: Electron pid=11, status<0
and $-11 < v_z < 0$ cm

Photons: pid =22, $E > 0.2$ GeV,

Use a 2σ cut: $29.12 - 2 * 10.52$
deg

RG-D study: π^0 analysis



I aimed to investigate the influence of the photon energy cut and the sector cut—where both photons must reside within the same sector—on the invariant mass.

Notably, the photon energy cut eliminates many photons.

Blue, ($E > 0.6$ GeV, sector cut, no cut on opening angle)

Red, ($E > 0.2$ GeV, sector cut, no cut on opening angle)

Black, ($E > 0.2$ GeV, no sector cut, cut on opening angle)

Green, ($E > 0.6$ GeV, no sector cut, no cut on opening angle)

Orange, ($E > 0.6$ GeV, no sector cut, cut on opening angle)

Red -> Background: 123580, Signal: 387946, S/B: 3.13923 diff: 264366

Blue -> Background: 10882, Signal: 101918, S/B: 9.36574 diff: 91036

Green -> Background: 17936, Signal: 110780, S/B: 6.1764 diff: 92844

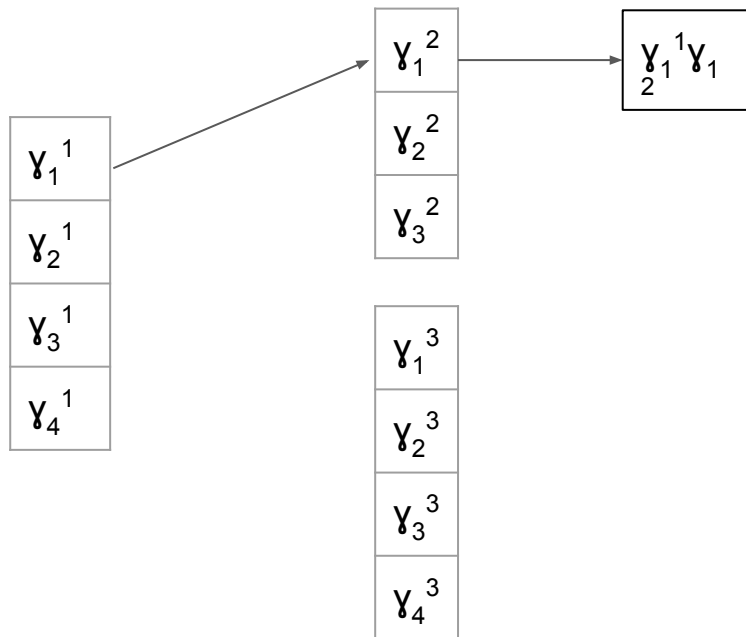
Orange -> Background: 11962, Signal: 104764, S/B: 8.75807 diff: 92802

Black -> Background: 137886, Signal: 427952, S/B: 3.10367 diff: 290066

RG-D study: π^0 analysis

Event mixing method:

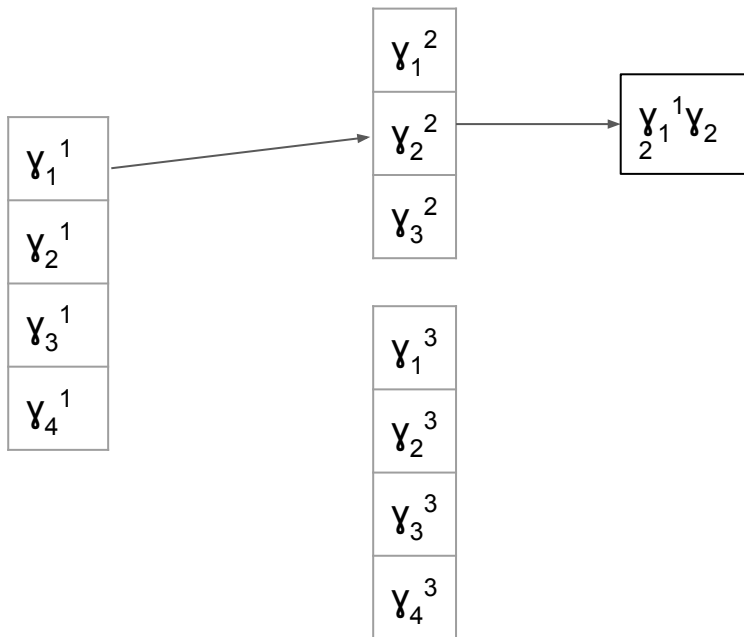
Implement using the C++ HIPO library. For that, I developed a function to go to the previous event. (It has been much more complicated than I was expecting).



RG-D study: π^0 analysis

Event mixing method:

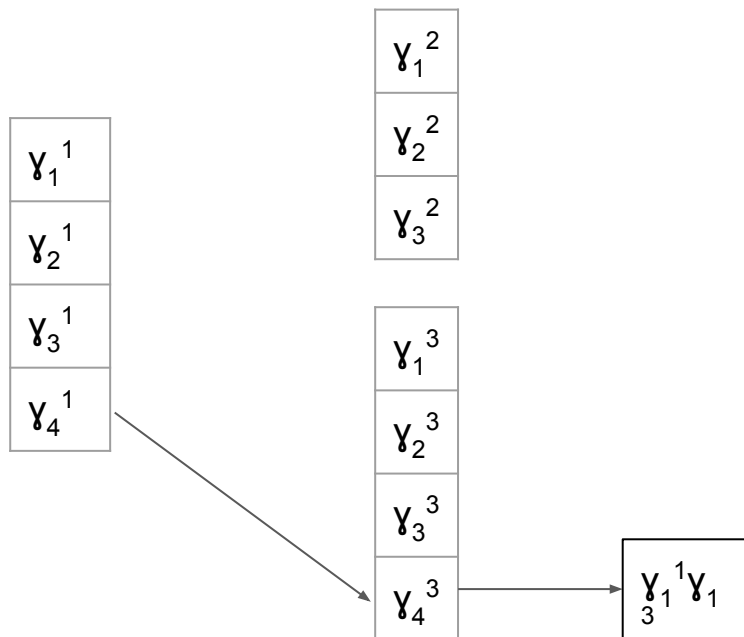
Implement using the C++ HIPO library. For that, I developed a function to go to the previous event. (It has been much more complicated than I was expecting).



RG-D study: π^0 analysis

Event mixing method:

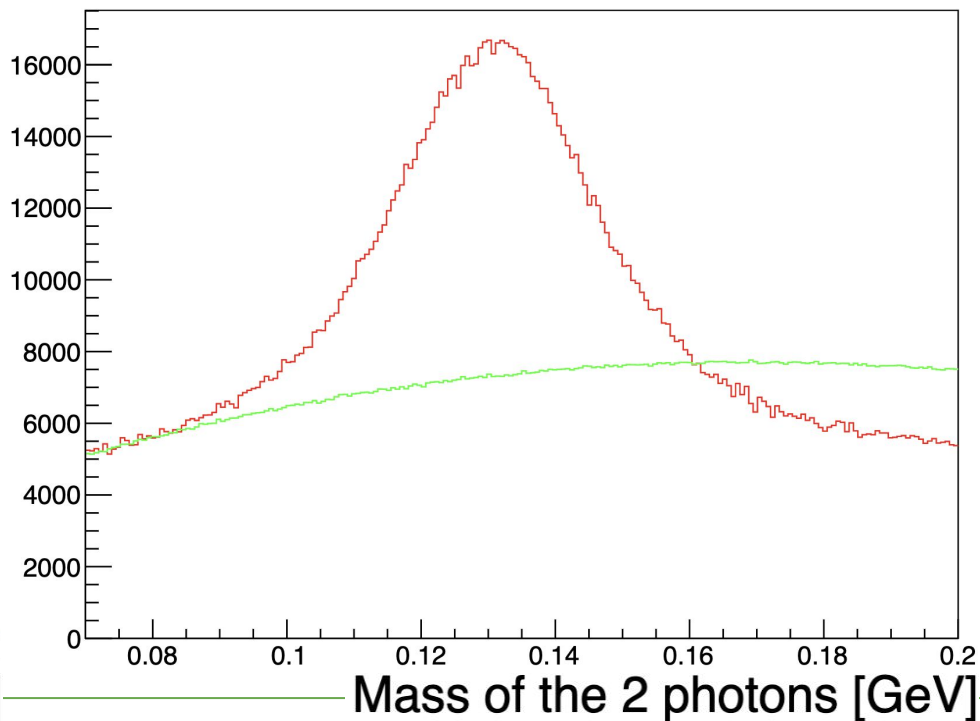
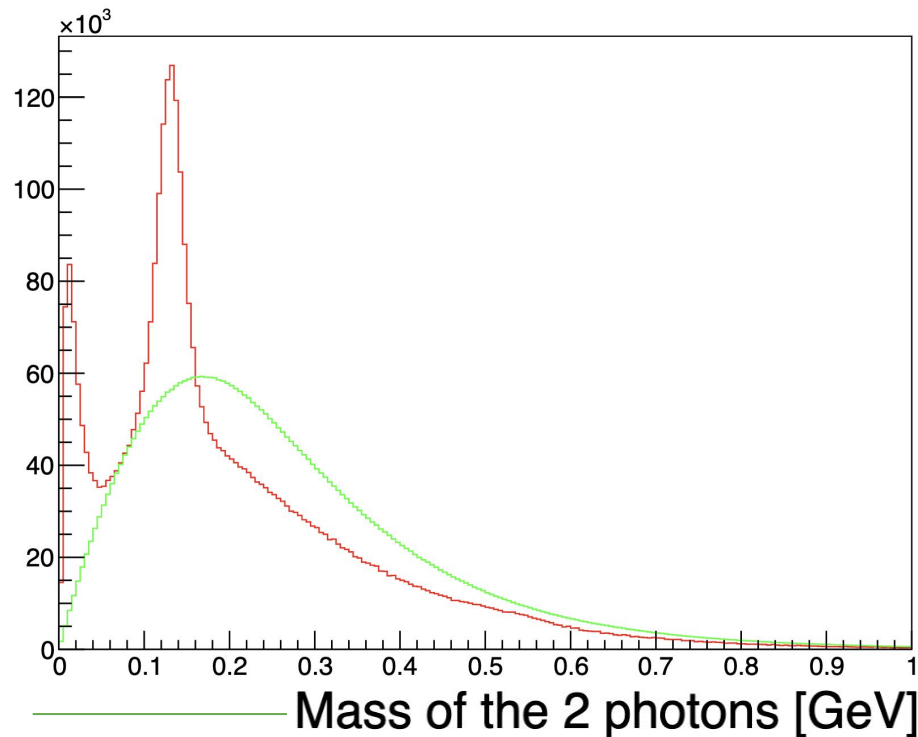
Implement using the C++ HIPO library. For that, I developed a function to go to the previous event. (It has been much more complicated than I was expecting).



At the end I got:

$$\begin{aligned}
 &Y_1^1 Y_1^2, Y_1^1 Y_2^2, & Y_4^1 Y_1^2, Y_4^1 Y_2^2, \\
 &Y_1^1 Y_3^2, & Y_4^1 Y_3^2 \\
 &Y_1^1 Y_1^3, Y_1^1 Y_2^3, & Y_4^1 Y_1^3, Y_4^1 Y_2^3, \\
 &Y_1^1 Y_3^3, & Y_4^1 Y_3^3, \\
 &Y_2^1 Y_1^2, Y_2^1 Y_2^2, \\
 &Y_2^1 Y_3^2 \\
 &Y_2^1 Y_1^3, Y_2^1 Y_2^3, \\
 &Y_2^1 Y_3^3
 \end{aligned}$$

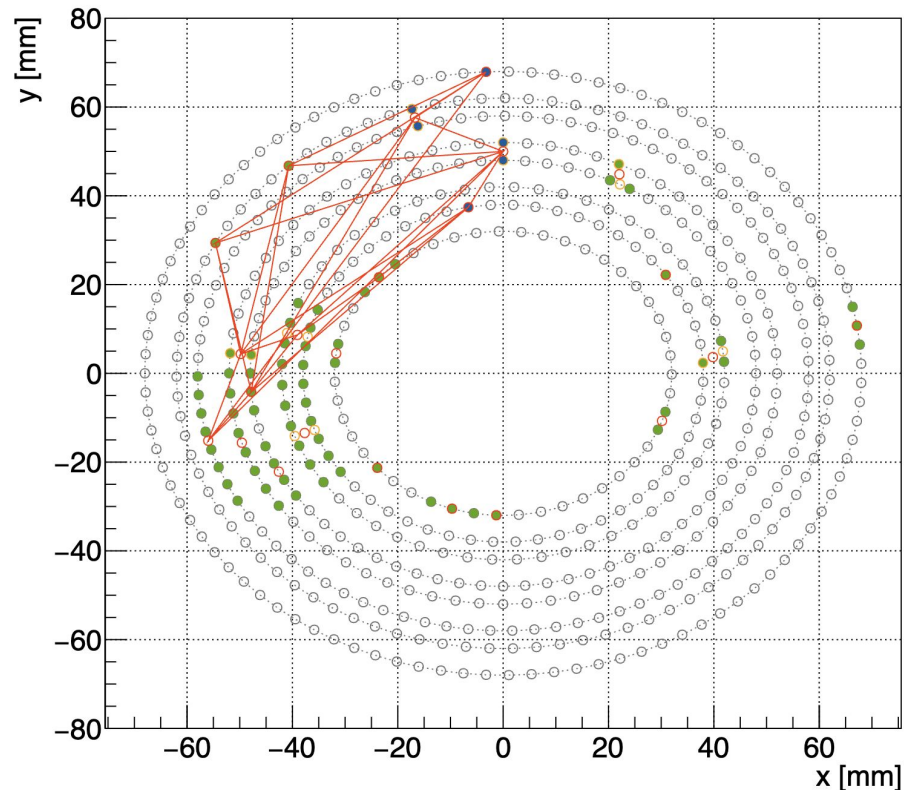
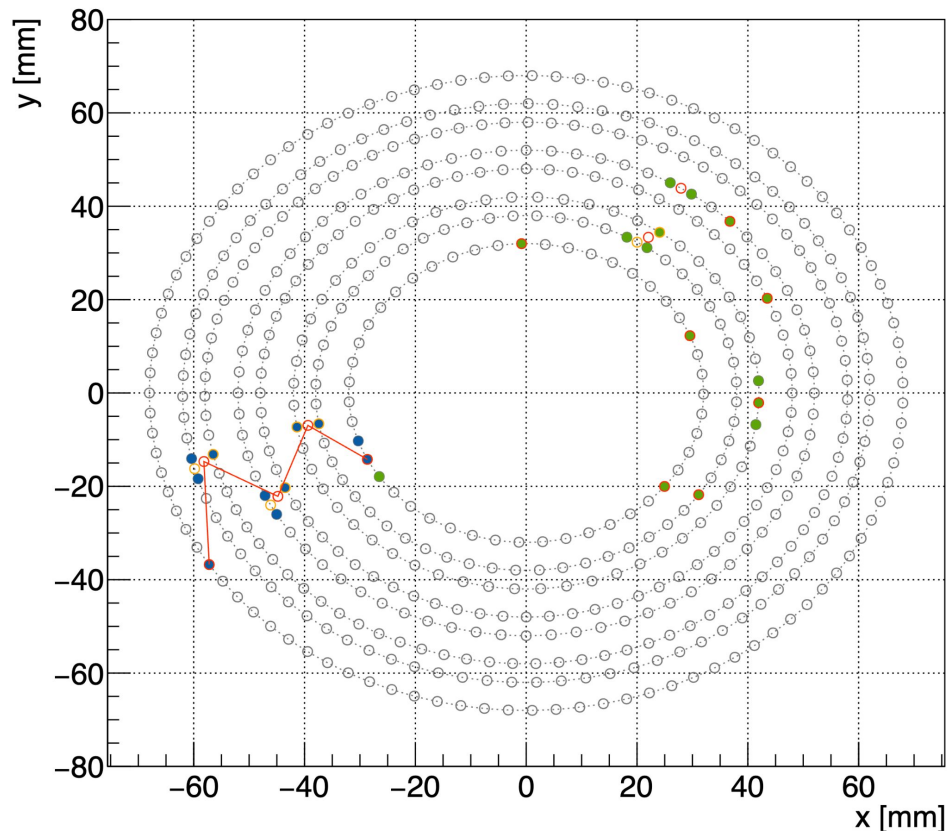
I iterate over the next 10 events; however, if fewer than ten events are available,
I iterate over the previous 10 events instead.



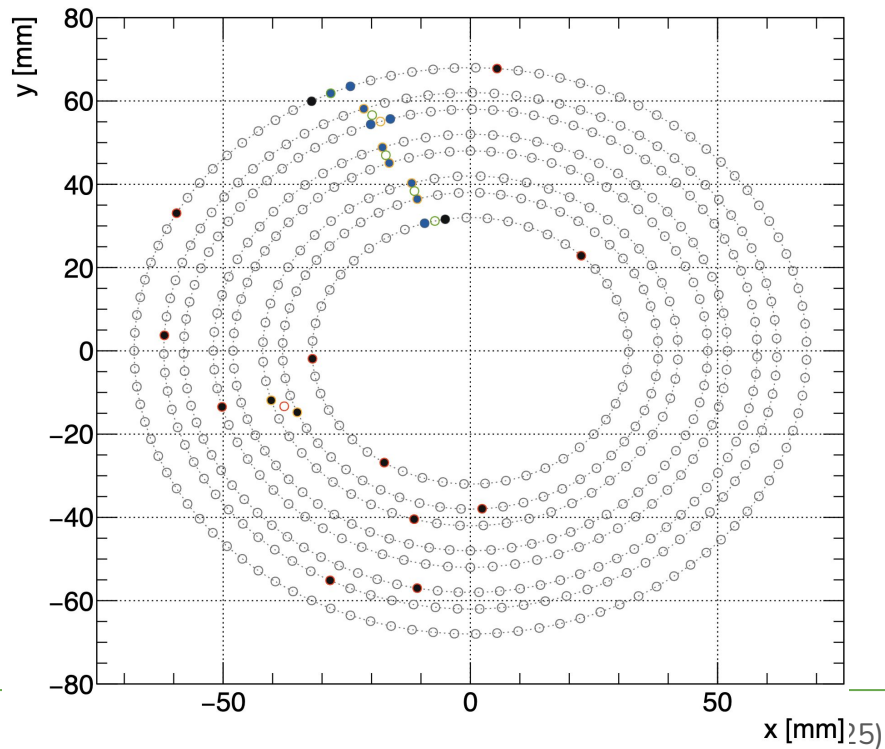
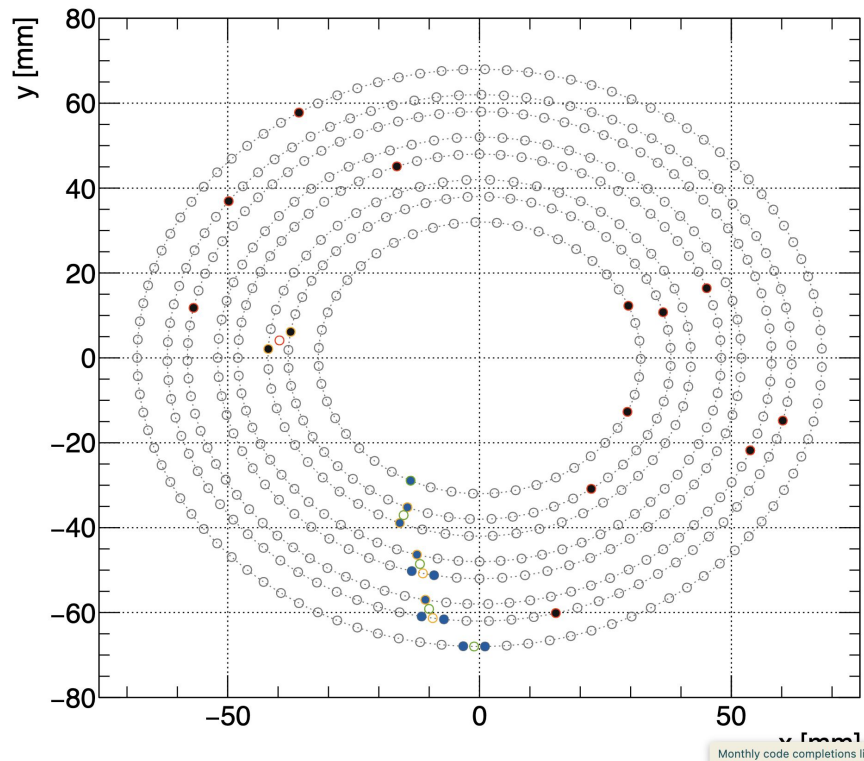
The event mixing method doesn't reproduce the background.

I need to find where is the issue.

Goal: use luminosity simulation as background. 2 events from the same HIPO file.



Using the previous method: randomly choose a wire.



- RG-D analysis:
 - Cross check for the nuclear transparency vs Q^2
 - Work on π^0 analysis:
 - Improve π^0 opening angle between the emitting scattered electron and the photon as function of the momentum ([Analysis note](#) by Stefan Diehl, Andrey Kim, Kyungseon Joo)
 - Fix the event mixing method
- Background merging, generate 100k events for one configuration
- AI for ALERT:
 - Redo the efficiency/purity analysis using luminosity simulation to have a better background
 - Comparison between conventional and AI track finding
- Question about the Collaboration meeting: I assume there will be a presentation on RGD. Would you like me to do it?